ST MARY'S DSG, KLOOF

GRADE: 10 **8 September 2023**

PHYSICAL SCIENCE CYCLE TEST
KINEMATICS

TIME: 55 MINS (incl. reading time) **TOTAL: 63**
ASSESSOR: T. Kirsten **MODERATORS: K. Bolton and G. Main**

NAME:_____

SCIENCE TEACHER:_____

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 7 pages. A data sheet is provided.
2. Answer on the question paper in the spaces.
3. Write your name and your science teacher's name on your question paper.

<u>QUESTION 1</u> – MULTIPLE CHOICE - **Choose the letter that you consider to be most correct and write your choice in the block alongside the question.**

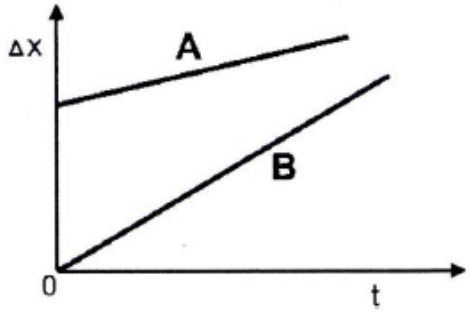
1.1 A student wanted to calculate the final velocity of a toy car, starting from rest, accelerating at a rate of 0,2 $m.s^{-2}$, and moving for 10 s. The formula the student used was:

$$v_f = v_i + a\Delta t$$

How many vector and scalar quantities does this formula contain?

	VECTOR QUANTITIES	SCALAR QUANTITIES
A	2	2
B	3	1
C	1	3
D	4	0

1.2 The motion of two objects, A and B, is represented in the following displacement-time graph.



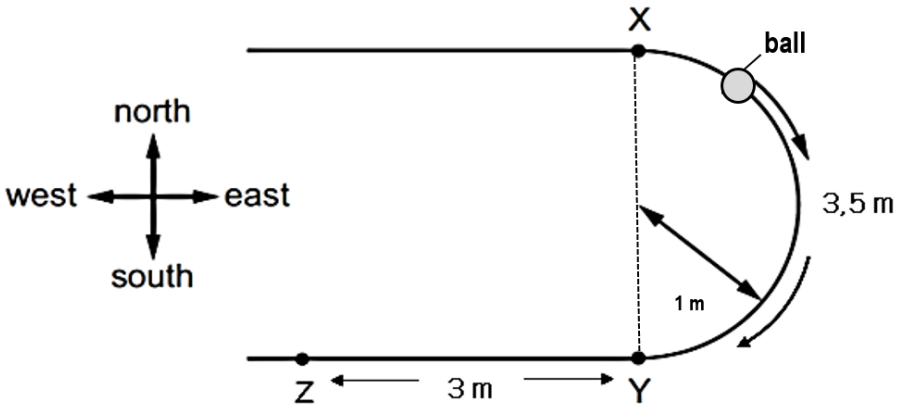
The graph indicates that:

A B is moving faster than A.
B A and B have different accelerations.
C A and B are on a collision course.
D A and B are moving in opposite directions.

2 x 2 = [4]

QUESTION 2

A ball travels from Point X to Point Y, a distance of 3,5 m, at a CONSTANT SPEED along a horizontal semi-circular track of **radius** 1 m in 2 s, as shown below.



2.1 What is the displacement of the ball from X to Y? (2)

_____

2.2 Define *velocity*. (2)

_____

2.3 Calculate the average velocity of the ball between X and Y. (3)

2.4 How would the average speed of the ball between X and Y differ to your answer to Q2.3? Write only MORE THAN, LESS THAN or EQUAL. (1)

_____

2.5 From Point Y, the ball then travels to Point Z, a distance of 3 m. Calculate the magnitude of the resultant displacement of the ball from the starting point (X). A sketch vector diagram must be included in your answer. (3)

[11]

QUESTION 3

3.1 A cyclist, heading **East** at a constant velocity of 9 $m.s^{-1}$, approaches the beach on a straight tar road.

- Soon the road changes into a soft gravel road. She enters the soft gravel road, and slows down to a velocity of 3 $m.s^{-1}$.
- This takes place over a time of 4 s on the soft gravel road.

3.1.1 Define *acceleration*. (2)

_____

3.1.2 Calculate the acceleration of the cyclist on the soft gravel road. (4)

3.2 The cyclist, travelling at the new velocity of 3 $m.s^{-1}$ on the gravel road, sees a rock in the road 5 m ahead. She applies her brakes and slows down at a rate of 1 $m.s^{-2}$ before stopping. Show by calculation if the cyclist will hit the rock or not. (4)

[10]

QUESTION 4

The table below shows the minimum stopping distances required for a car travelling at various speeds on a dry road with good brakes and good tyres. The car is travelling **North**.

Initial speed of car ($km.h^{-1}$)	**Distance travelled during reaction time (m)**	**Distance travelled while braking (m)**
32	6	6
48	9	14
64	12	24
80	15	38
97	16	58

4.1 Describe the relationship between minimum stopping distance for a car and the initial speed of the car. (2)

_____

_____

4.2 Convert the speed of 97 $km.h^{-1}$ into $m.s^{-1}$. (1)

4.3 Calculate the reaction time of the driver in the car travelling at **97 $km.h^{-1}$**. (3)

4.4 List 2 factors which could increase the reaction time of the driver**.** (2)

_____

_____

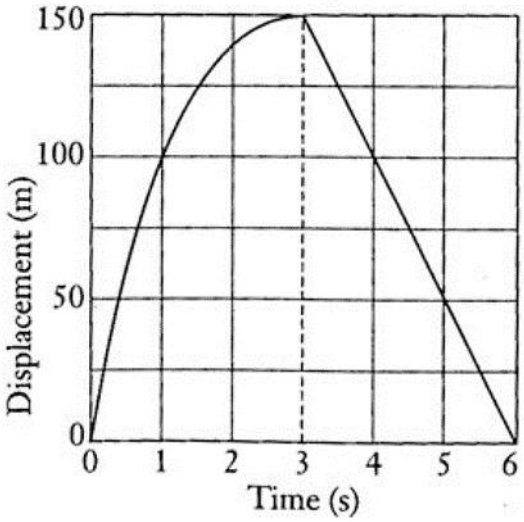
4.5 What is the distance travelled during the driver's reaction time known as? (1)

_____

[9]

QUESTION 5

The displacement-time graph below represents the motion of a motorbike along a straight race track, **West**.



5.1 Describe the motion of the motor bike:

i) from 0 - 3 s. (2)

_____

ii) from 3 - 6 s. (2)

_____

5.2 What is the displacement of the motor bike from the starting point after 6 s? (1)

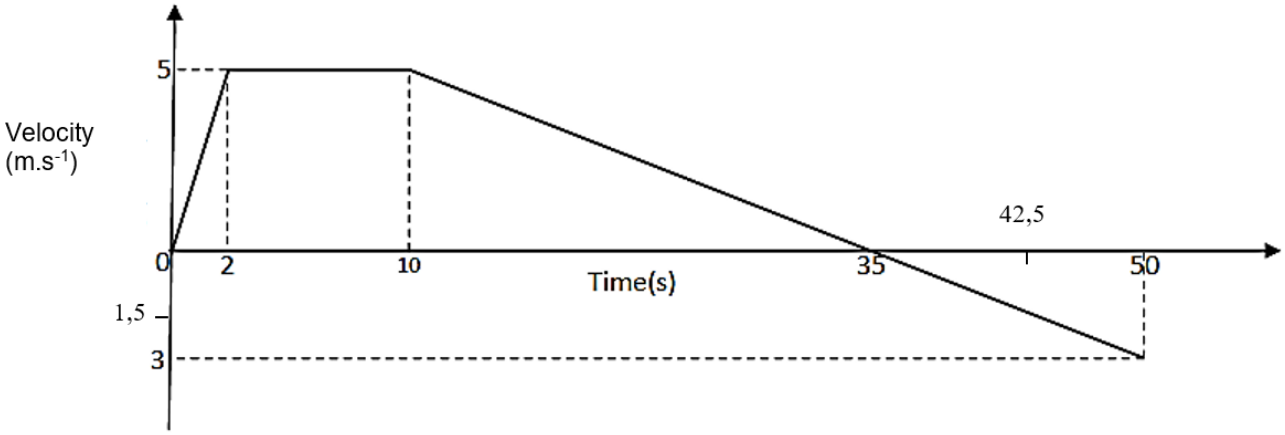
_____

[5]

QUESTION 6



Gerda Steyn is a South African marathon and ultramarathon athlete. Gerda sets out on a training run. She starts off by running in a **Southerly** direction. Her motion during the first part of her run is graphed below.



6.1 Using the graph, calculate Gerda's acceleration during the first two seconds of the run. (4)

6.2 Calculate Gerda's average velocity in the first 10 s of her run. (4)

6.3 Describe Gerda's motion between 10 and 50 s. (6)

_____

_____

_____

6.4 What is Gerda's instantaneous velocity at 42,5 s? (2)

_____

6.5 Calculate Gerda's displacement between 10 and 50 s (4)

6.6 On the axes below, sketch an acceleration-time graph that represents the first 50 s of Gerda's run. You need only show actual values on the time axis. It is the shape that is important. (4)



[24]